

Final Year Project

Foundations Report

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Abstract

This project investigated how Natural Language Processing (NLP) can be used to automatically extract and structure data quality metric definitions from academic papers, standards documents, and table-heavy research sources. Metric definitions are typically dispersed across heterogeneous documents, making them difficult to discover, compare, and reuse in practice.

The proposed solution is an end-to-end extraction pipeline that identifies candidate metric definitions, extracts key fields (e.g., metric label, definition, formula, and dimension), and represents the results as Resource Description Framework(RDF) aligned with the World Wide Web Consortium (W3C) Data Quality Vocabulary (DQV), Simple Knowledge Organization System (SKOS), and Dublin Core. The extracted data is represented in RDF and queried using SPARQL and RDF Query Language (SPARQL), both of which are standard Semantic Web technologies [1, 2].

The project demonstrates the feasibility of automating data quality metric curation and provides a practical alternative to fully manual knowledge collection.

Declaration on Generative AI. I declare that I used ChatGPT as a supportive aid during the preparation of this report for grammar checking, improving clarity and readability, resolving LaTeX formatting issues, and assisting with the presentation of diagrams and tables. The tool was not used to generate original research ideas, data, analyses, results, or final conclusions. All research design, implementation, evaluation, interpretation of results, and final academic arguments are my own responsibility.

Chapter 1: Project Description

1.1 Problem Context

Data quality is commonly discussed through high-level dimensions such as completeness, accuracy, and consistency. However, the concrete metrics that operationalise these dimensions are dispersed across academic papers and standards documents, typically presented in unstructured or semi-structured formats.

This fragmentation creates several practical challenges. First, discovering relevant metrics requires manual search across heterogeneous sources. Second, comparing definitions across documents is difficult due to inconsistent terminology and formatting. Third, formal metric definitions, including formulas, are not readily reusable in machine-readable form. As a result, metric knowledge is often duplicated, inconsistently interpreted, and difficult to integrate into data-driven systems.

These limitations highlight the need for automated methods to extract, standardise, and structure data quality metric definitions in a form that supports reuse, comparison, and systematic querying.

1.2 Project Aim

The aim of this project is to design and evaluate a hybrid extraction pipeline that automatically identifies and structures data quality metric definitions from academic papers, standards documents, and table-heavy sources, and represents them in a standards-aligned, machine-readable knowledge graph.

The project focuses not only on extraction accuracy, but also on the usefulness of the resulting structured representation (the knowledge graph) for downstream querying and knowledge discovery.

1.3 Research Questions

This project is guided by the following research questions:

- **RQ1 (Extraction Performance):** To what extent can a hybrid extraction pipeline accurately extract data quality metrics from heterogeneous document types?
- **RQ2 (Structured Representation):** To what extent does representing extracted metrics in a knowledge graph enable meaningful structured retrieval of data quality metric knowledge?

In this context, meaningful structured retrieval refers to the ability to query the knowledge graph for information needs relevant to metric discovery and reuse, including retrieving metrics by source,

navigating metrics by quality dimension, and accessing detailed metric properties such as labels, definitions, and formulas.

1.4 Expected Artefacts

The intended deliverables are:

- A hybrid extraction pipeline for identifying metric definitions and associated fields in academic papers, standards documents, and table-based research sources.
- A knowledge graph containing extracted metric entities aligned with W3C vocabularies (e.g., DQV, SKOS, Dublin Core).
- A data quality metric knowledge base consisting of the generated RDF data loaded into a GraphDB triple store, with SPARQL used as the query interface.
- A manually annotated gold-standard dataset for evaluation.
- A set of SPARQL queries demonstrating structured retrieval over the extracted metrics.

1.5 Success Criteria

Success will be assessed through:

- **Extraction quality:** precision, recall, and F1-score against a gold standard.
- **Property-Level accuracy:** correctness of extracted metric properties (label, definition, formula, dimension).
- **Record completeness:** proportion of correctly extracted fields per metric.
- **Queryability:** ability to retrieve and explore metrics using SPARQL queries aligned with representative user tasks.

1.6 Baseline System Architecture

The agreed baseline implementation follows a modular pipeline:

*PDF/Text → Python extraction → Structured output (CSV) → R2RML mapping →
RDF/Turtle → GraphDB.*

This modular design separates extraction, transformation, and storage concerns, allowing each stage to be improved independently. In particular, it enables iterative refinement of the extraction component without requiring changes to the knowledge graph representation or query layer.

Chapter 2: Introduction

Data quality metrics play a central role in data management, analytics, and machine learning by providing measurable indicators of whether data is suitable for a given purpose. While high-level data quality dimensions such as completeness, accuracy, and consistency are widely recognised [3], the concrete metrics that operationalise these dimensions are often dispersed across academic papers and standards documents [3–5]. These sources are typically presented in unstructured or semi-structured formats and are designed for human interpretation rather than machine reuse.

This fragmentation creates a number of practical challenges. Discovering relevant metrics requires manual search across heterogeneous documents, while comparing definitions and formulas across sources is time-consuming and error-prone. Furthermore, the lack of structured representation prevents systematic querying, linking, and reuse of metric knowledge. As a result, practitioners must repeatedly reconstruct knowledge that already exists, leading to inefficiencies and inconsistencies in how data quality is assessed.

This project addresses these challenges by investigating whether data quality metric definitions can be automatically extracted from text and represented in a structured, machine-readable form. The work is guided by two research questions: (i) to what extent a hybrid extraction pipeline can accurately extract metrics from heterogeneous document types, and (ii) whether representing these metrics in a knowledge graph enables meaningful structured retrieval of metric knowledge.

To answer these questions, the project proposes a hybrid extraction approach that combines rule-based methods, table-specific reconstruction, and NLP-based techniques to handle different document structures. Extracted metrics are transformed into a structured representation and mapped to a knowledge graph aligned with established semantic vocabularies. The model uses Data Quality Vocabulary for representing metrics and dimensions [6], SKOS for concept modelling and taxonomy organisation [7], and Dublin Core metadata terms for provenance and source information [8].

Using established vocabularies provides shared semantics for the extracted entities and relationships, making the resulting knowledge graph easier to interpret, query, and extend. It also improves interoperability, since metrics, dimensions, labels, definitions, and provenance can be represented using commonly understood RDF terms rather than project-specific fields.

This enables the resulting data to be queried using SPARQL and supports structured exploration of metric knowledge by source, dimension, and metric detail.

The system is evaluated using a manually constructed gold-standard dataset containing 123 annotated metrics across multiple source types. Evaluation focuses on extraction performance using precision, recall, and F1-score, complemented by Property-Level accuracy, record completeness, and partial match analysis to better capture practical usefulness. In addition, representative SPARQL queries are used to demonstrate that the resulting knowledge graph supports meaningful retrieval tasks.

The remainder of this report is structured as follows. Chapter 3 reviews related work across data quality metrics, knowledge graphs, and information extraction. Chapter 4 describes the data sources and gold standard construction. Chapter 5 outlines the system design and implementation. Chapter 6 presents the evaluation. Chapter 7 describes the project workplace, and Chapter 8 summarises conclusions and future work.

Chapter 3: Related Work and Ideas

This chapter reviews work across three strands that directly inform the project: (i) data quality metrics, (ii) knowledge graphs and semantic data representation, and (iii) information extraction from academic papers, standards documents, and table-heavy sources. These strands motivate the design of the proposed system and highlight the gap addressed by this work.

3.1 Data Quality Metrics

Data quality has been extensively studied, with surveys such as Zaveri et al. [3] providing comprehensive overviews of quality dimensions and associated metrics. These works organise metrics under conceptual categories such as completeness, accuracy, and consistency, and provide examples of how such metrics are defined and applied.

However, the state of the art keeps moving forward and new metrics are continuously proposed, meaning that no single survey or standard can be considered exhaustive. This further motivates the need for automated extraction systems capable of ingesting new sources as they emerge.

Similarly, standards bodies such as ETSI¹ define formalised approaches to data quality measurement, often including detailed descriptions and formulas for metrics. These sources provide authoritative definitions but are typically distributed as structured documents intended for human interpretation rather than machine processing.

Despite the richness of these resources, data quality metrics are rarely published in a structured, queryable format. Instead, they are embedded within narrative text or tables, making them difficult to discover, compare, and reuse. This limitation motivates the need for automated extraction and structuring of metric knowledge.

3.2 Knowledge Graphs and Semantic Representation

Knowledge graphs provide a structured way to represent domain knowledge using entities, relationships, and shared vocabularies [9]. They enable expressive querying and integration across heterogeneous data sources [9]. In the context of data quality, the W3C DQV [6] defines a standard model for representing metrics, dimensions, and quality assessments.

DQV is often used in conjunction with other semantic web standards such as SKOS [7] for concept modelling and Dublin Core [8] for provenance and metadata. Together, these vocabularies enable the construction of interoperable knowledge graphs that can support structured querying via SPARQL.

¹<https://www.etsi.org/technical-groups/data/>

R2RML is also relevant to this project because it provides a standard way to map structured tabular or relational data into RDF. In this pipeline, extracted metric records are first stored in CSV format as an intermediate structured representation, then mapped into RDF using an R2RML mapping. This separates the extraction stage from the semantic representation stage, allowing the extracted data to be inspected before conversion into knowledge graph triples [10].

However, most knowledge graph applications assume that structured data is already available. They do not address the problem of extracting structured representations from unstructured technical documents, which remains a key challenge in this project.

3.3 Information Extraction from Unstructured Text

Information extraction techniques aim to convert unstructured text into structured representations [11]. Common information extraction tasks include named entity recognition, relation extraction, and attribute extraction. These tasks can be approached using rule-based methods, pattern matching, statistical models, or more recent machine learning and transformer-based techniques.

A key challenge in information extraction is maintaining precision while still capturing enough relevant information. Rule-based approaches can achieve high precision when documents follow predictable patterns, but they may fail when text structure varies. More flexible machine learning and transformer-based approaches can generalize across diverse inputs, but this often comes with increased computational complexity, greater data requirements, and reduced explainability.

In this project, the emphasis is on rule-based and hybrid extraction rather than supervised machine learning. This design choice supports explainability and controllability, which are important when extracting structured metric definitions from technical documents where errors need to be inspected and traced.

Hybrid approaches combining rule-based and NLP-based methods can balance precision and flexibility in technical domains [12]. This motivates the design of the extraction pipeline proposed in this work, where structured extraction is used for standards-style documents, table-specific reconstruction is used for survey sources, and NLP-based processing is used as a fallback for less regular text.

Extracting data quality metrics introduces additional challenges compared to basic named entity recognition [11]. Metric definitions may span multiple sentences, appear within tables or structured sections, and include formulas or mathematical expressions. PDF-based sources further complicate extraction due to formatting artifacts and loss of structural information, particularly where table layouts are flattened during text extraction [13].

These challenges motivate the use of a hybrid extraction pipeline that combines rule-based extraction for structured content, custom parsing logic for flattened table-based sources, and spaCy-based sentence segmentation and dependency parsing for less structured text, supported by regular-expression fallback rules.

Recent work on knowledge graph construction from unstructured text has emphasized the importance of systematic evaluation against gold-standard annotations, often using metrics such as precision, recall, and F1-score [14]. Such approaches highlight that multi-level evaluation — combining record-level metrics with property-level and partial match analysis — provides a more complete picture of extraction quality than binary correctness alone, particularly when outputs may be semantically correct but textually good.

3.4 Gap and Motivation

The reviewed literature shows that data quality metrics, semantic vocabularies, and information extraction techniques each address part of the problem considered in this project. Data quality literature provides metric definitions and taxonomies, while semantic web standards provide a way to represent structured knowledge. Information extraction methods provide techniques for converting text into structured outputs.

However, the structure of the source documents makes a single extraction method insufficient. Standards-style documents, survey tables, and semi-structured research papers present metric definitions in different forms. This motivates a hybrid extraction approach, where deterministic rules can be used for predictable document sections, custom parsing logic can be used for flattened table-based sources, and NLP-based processing can support less regular text. The importance of hybrid methods is therefore practical: they allow the extraction strategy to be adapted to the structure of each source while maintaining control and explainability.

In knowledge graph construction, extraction quality is commonly evaluated by comparing extracted entities and properties against a gold standard using measures such as precision, recall, and F1-score. For this project, this means evaluating whether each extracted `dqv:Metric` instance corresponds to a gold-standard metric, and whether its associated properties, such as label, definition, formula, and dimension, are correctly captured. This motivates a multi-level evaluation design that considers both record-level correctness and property-level accuracy.

Across these strands of research, a clear gap exists between (i) the availability of high-quality metric definitions in academic and standards-based sources, and (ii) their accessibility in a structured, machine-readable form.

Existing work provides strong conceptual foundations for data quality metrics and robust frameworks for representing structured knowledge, but does not bridge the gap between unstructured documentation and structured knowledge graphs of data quality metrics. To the best of my knowledge, there is no existing work that specifically addresses the automatic extraction of data quality metric definitions and their transformation into RDF representations aligned with standard vocabularies such as DQV, SKOS, and Dublin Core.

This project addresses this gap by combining a hybrid extraction pipeline with standards-aligned knowledge graph modelling. By transforming unstructured metric definitions into a queryable RDF representation, the project enables structured retrieval of metric knowledge through SPARQL queries, supporting improved discoverability, comparison, and reuse.

Chapter 4: Data Considerations

4.1 Data Sources

The gold standard dataset was constructed from three primary sources selected to represent different document structures commonly encountered in data quality literature:

- **ETSI**: a structured standards-style document with clearly defined metric descriptions.
- **Zaveri et al. (2015)**: a survey-style source where metrics are primarily presented in tabular form.
- **IEEE survey paper**: a semi-structured research document containing multiple tables and descriptive text.

These sources were deliberately chosen to introduce structural diversity into the dataset. The ETSI document provides a relatively regular format suitable for rule-based extraction, while the Zaveri and IEEE sources introduce increasing levels of structural ambiguity through table-based and semi-structured representations. This diversity enables evaluation of the extraction pipeline across realistic document types rather than a single controlled format.

4.2 Source Selection Criteria

The selected sources were chosen based on the following criteria:

- **Relevance**: Each source contains well-defined and widely referenced data quality metrics.
- **Structural diversity**: The sources differ significantly in how metrics are presented (structured definitions, tables, semi-structured text).
- **Coverage**: Together, the sources span multiple quality dimensions and provide a varied set of metric types.
- **Evaluation suitability**: The documents contain sufficient detail (e.g., labels, definitions, formulas) to support reliable manual annotation and comparison.

This selection strategy ensures that the evaluation reflects realistic challenges in extracting metric definitions from heterogeneous documents.

4.3 Gold Standard Construction

A total of 123 metrics were manually annotated across the selected sources to form the gold standard dataset used for evaluation. The gold standard was represented as a structured tabular dataset rather than RDF. Each row corresponded to one annotated metric instance, with columns matching the main fields produced by the extractor: MetricLabel, Definition, Formula, Dimension, and Source.

This tabular format was chosen because it allowed direct comparison between manually annotated records and the CSV output of the extraction pipeline. The extracted records were later transformed into RDF for knowledge graph construction, but the primary gold standard used for precision, recall, property-level accuracy, and partial match analysis remained in tabular form.

Metrics were selected systematically to ensure:

- full coverage of smaller sources (e.g., ETSI),
- representative coverage across larger tables (e.g., Zaveri and IEEE),
- inclusion of metrics with varying levels of structural complexity.

Each annotated record includes key fields such as metric label, definition, formula (where available), source, dimension, and metric type. The resulting dataset supports both record-level evaluation (precision, recall, F1) and property-level analysis. Figure 4.1 shows a representative example of an annotated gold-standard record in the tabular format used for evaluation.

Gold Standard Record Example (Zaveri)						
MetricID	MetricLabel	Definition	Formula	Source	Dimension	MetricType
high-throughput	high throughput	No. of answered HTTP-requests per second		.,NULL,Zaveri	performance	quantitative

Figure 4.1: Representative gold-standard record in tabular/CSV format. Each row corresponds to one annotated metric instance and contains the fields used for comparison with the extractor output.

It is important to note that some metrics appear multiple times within the same source (e.g., across different tables). In these cases, the gold standard preserves each occurrence, while the extraction pipeline may deduplicate records based on metric label. This difference in representation is considered during evaluation.

4.4 Normalisation

During gold standard construction, PDF extraction artefacts such as line-break hyphenation (e.g., “con-sidered”) and minor typographical inconsistencies were manually normalised to ensure fair evaluation. These artefacts arise from document formatting rather than genuine linguistic variation.

Without this normalisation step, evaluation metrics such as precision and recall would be artificially penalised by superficial textual differences rather than reflecting true extraction performance.

4.5 Challenges in Source Data

PDF-based documents introduce several challenges for automated extraction, particularly where table structure and layout information must be recovered from a format designed primarily for presentation rather than structured data extraction [13].

- **Loss of table structure:** Tabular layouts are often flattened into plain text, making it difficult to reconstruct row-based metric entries.
- **Broken text spans:** Sentences may be split across lines or merged incorrectly during extraction.
- **Hyphenation artefacts:** Words may be split across line breaks (e.g., “con-sidered”), introducing noise into extracted text.

These issues directly impact both extraction quality and evaluation, as they can lead to partial matches even when the correct metric has been identified.

4.6 Source-Specific Considerations

The selected sources differ not only in structure but also in how they conceptualise data quality metrics.

In particular, the IEEE source organises metrics according to economic value categories (e.g., utility, cost, market value) rather than standard data quality dimensions. As a result, dimension labels are not directly aligned with the DQV taxonomy used in the knowledge graph and must be inferred heuristically.

Similarly, table-based sources such as Zaveri require reconstruction of metric entries from partially structured text, introducing additional complexity compared to standards-style documents such as ETSI.

These differences highlight the importance of adopting a hybrid extraction strategy and are reflected in the evaluation results.

Chapter 5: Outline of Approach

5.1 Scope

The project focuses on building a knowledge base of data quality metrics by extracting metric information from academic papers and standards documents and representing it as a knowledge graph. The goal is not to perform data quality assessment on datasets, but rather to curate and expose reusable metric knowledge, including metric definitions, dimensions, categories, and provenance.

This distinction is important, as the system targets knowledge extraction and structuring rather than data evaluation. The resulting knowledge graph is intended to support discovery, comparison, and reuse of metric definitions through structured querying.

5.2 System Overview

The system follows a modular end-to-end pipeline that separates extraction, transformation, and storage:

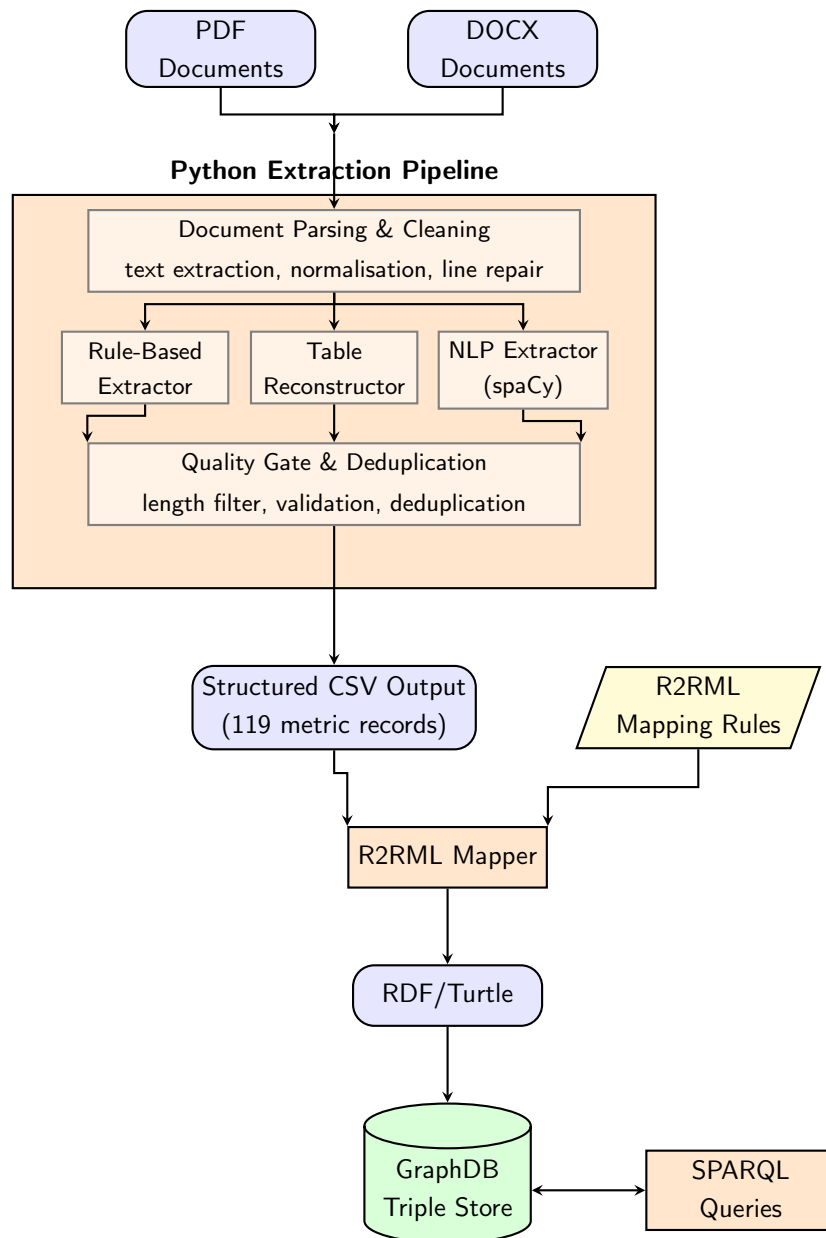


Figure 5.1: End-to-end system pipeline from document ingestion to RDF generation and GraphDB querying. Rounded rectangles represent input/output data artefacts, plain rectangles represent processing components, the trapezoid represents the R2RML mapping rules, and the cylinder represents the GraphDB triple store.

This design enables incremental development, where improvements to the extraction stage can be made independently of the knowledge graph representation. It also ensures that intermediate outputs (e.g., CSV files) can be inspected and validated before transformation into RDF.

5.3 Pipeline Design

The pipeline consists of four main stages:

- **Document ingestion:** Input documents (PDF or text) are parsed and preprocessed, including basic cleaning such as whitespace normalisation and line reconstruction.
- **Metric extraction:** Candidate metric definitions are identified and structured fields (label, definition, formula, dimension) are extracted using a hybrid approach.
- **Structured representation:** Extracted data is stored in a tabular format (CSV), providing a transparent intermediate representation.
- **Knowledge graph transformation:** The structured data is mapped to RDF using R2RML, following the W3C R2RML mapping standard [10], and loaded into GraphDB for querying.

This staged approach improves traceability and allows errors to be isolated to specific parts of the pipeline.

5.4 Hybrid Extraction Strategy

The extraction pipeline adopts a hybrid design combining multiple strategies to handle the structural diversity of input documents.

- **Structured extraction:** For standards-style documents (e.g., ETSI), rule-based extraction is used to identify metric definitions and associated fields. These documents typically follow predictable patterns, allowing high-precision extraction.
- **Table-based extraction:** For survey-style documents (e.g., Zaveri, IEEE), structural patterns and row anchors are used to reconstruct metric entries from flattened table text. This is necessary due to the loss of explicit table structure during PDF extraction.
- **NLP-based extraction:** For unstructured text, spaCy is used for sentence segmentation and dependency parsing, supported by regex-based fallback methods. This allows the system to handle less predictable document layouts.

Algorithm 1 Hybrid Metric Extraction

```

1: for each document do
2:   Parse document (PDF/DOCX)
3:   if document is structured (DOCX) then
4:     Apply rule-based extraction
5:   else if document contains tables then
6:     Apply table reconstruction
7:   else
8:     Apply NLP-based extraction (spaCy)
9:   end if
10:  for each candidate metric do
11:    if definition length  $\geq$  threshold then
12:      Extract (label, definition, formula, dimension)
13:      Deduplicate by (label, source)
14:      Write to CSV
15:    end if
16:  end for
17: end for

```

This hybrid approach reflects the varying structure of source documents and balances precision and flexibility. A single uniform extraction strategy would not perform well across all input types.

5.5 Implementation Details

The extraction component is implemented in Python using a combination of rule-based logic and NLP tooling. spaCy is used for sentence segmentation and basic linguistic processing [15], while regular expressions are used for pattern matching and fallback extraction. Document parsing was performed using PyMuPDF [16] for PDF text extraction and preprocessing.

Example Extraction

Table 5.1 shows a concrete example of how an ETSI metric definition is transformed through the pipeline, from source text to CSV output and RDF representation.

Stage	Content
Input (source text)	"Attribute Completeness: The completeness of a specific attribute A_j is given by the ratio of present values for that attribute to the total number of records. Formula: $C_{A_j} = \frac{\sum_{i=1}^N I_{ij}}{N}$"
Extracted CSV	MetricID: attribute-completeness, MetricLabel: Attribute Completeness, Definition: The completeness of a specific attribute A_j is given by the ratio of present values for that attribute to the total number of records., Formula: C_Aj = (sum(i=1 to N) I_ij)/N, Source: ETSI, Dimension: completeness, MetricType: quantitative
RDF representation	<pre> ex:attribute-completeness a skos:Concept, dqv:Metric ; skos:prefLabel "Attribute Completeness"@en ; skos:definition "The completeness of a specific attribute A_j is given by the ratio of present values for that attribute to the total number of records."@en ; skos:inScheme ex: ; skos:note "Extracted from ETSI"@en ; dqv:inDimension ldqd:completeness ; ex:formula "C_Aj = (sum(i=1 to N) I_ij)/N" ; ex:metricType ex:metricType/quantitative ; dcterms:bibliographicCitation "ETSI" . </pre>

Table 5.1: Example of metric extraction from source text to RDF representation.

A key design constraint was to prioritise explainability and controllability of the extraction process. As a result, the system avoids complex machine learning models and instead relies on deterministic rules and heuristics that can be inspected and adjusted.

Post-processing steps include filtering low-quality candidates, enforcing minimum definition length, and simple deduplication based on metric label and definition.

5.6 Knowledge Graph Model

The knowledge graph model is based on W3C-aligned vocabularies to ensure interoperability and semantic clarity.

- **Core concepts:** `dqv:Metric`, `dqv:Dimension`, `dqv:Category`
- **Concept modelling:** Dimensions and categories are represented as `skos:Concept` to support hierarchical organisation
- **Relationships:** `dqv:inDimension` and `dqv:inCategory` link metrics to their respective concepts
- **Labels and definitions:** `rdfs:label` and `skos:definition` store human-readable information
- **Provenance:** `dct:source` and `dcterms:bibliographicCitation` capture source information

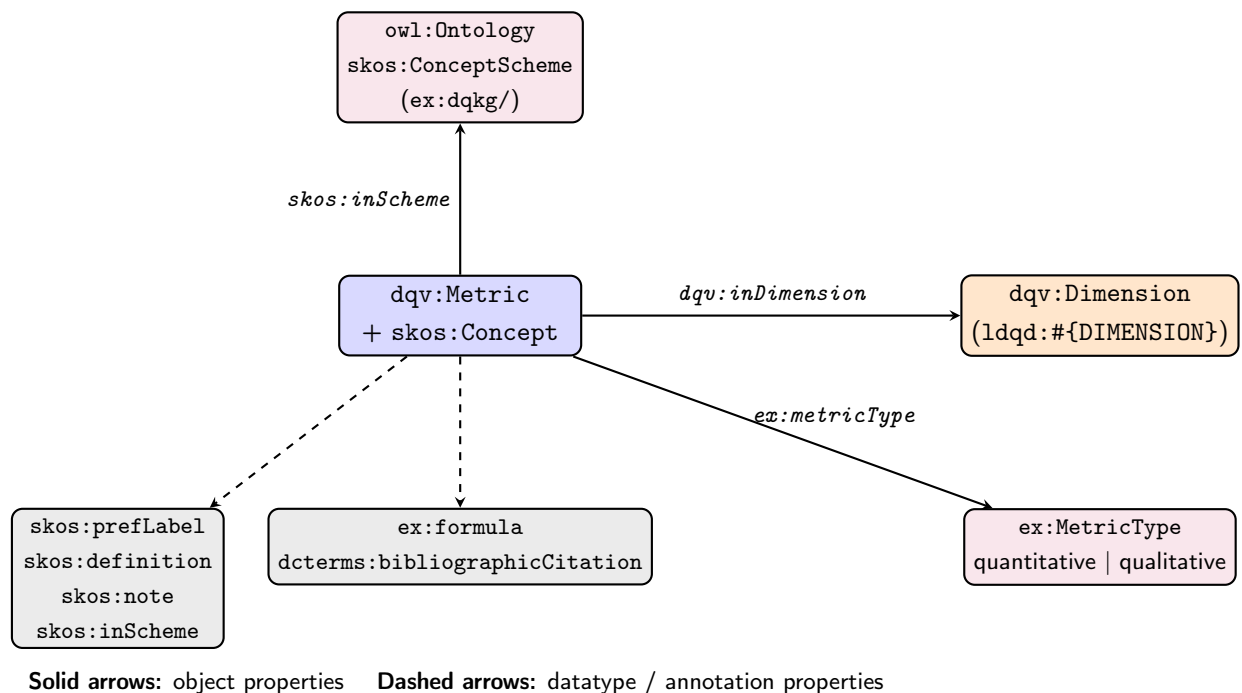


Figure 5.2: Knowledge graph model derived from the R2RML mapping.

This model enables structured querying of metrics and supports integration with existing semantic web tools.

5.7 Design Considerations and Limitations

Several design decisions introduce trade-offs:

-
- **PDF dependency:** Extraction quality is limited by the reliability of PDF text extraction, particularly for tables and formulas.
 - **Heuristic dimension inference:** Dimensions are inferred using keyword-based heuristics rather than authoritative labels, which may lead to semantically correct but non-identical classifications.
 - **Source variability:** Dimensions are inferred using keyword-based heuristics rather than authoritative labels. For sources using non-DQV taxonomies (e.g., IEEE economic value categories), the extractor outputs `unknown` rather than attempting an incorrect mapping. This is a known limitation of the current implementation.

These limitations are reflected in the evaluation results and highlight areas for future improvement.

Chapter 6: Evaluation

This chapter evaluates the effectiveness of the proposed extraction pipeline. It first assesses extraction performance using precision, recall, and F1-score under both exact and lenient matching criteria. It then examines Property-Level accuracy and record completeness, followed by a qualitative analysis of partial matches and error categories. Finally, the chapter demonstrates the queryability of the resulting knowledge graph using SPARQL and provides a comparative baseline using large language models.

Together, these analyses provide a comprehensive evaluation of both the correctness and practical usefulness of the extracted data quality metrics.

6.1 Evaluation Objectives and Hypotheses

The evaluation was designed to assess the effectiveness of the proposed extraction pipeline across source documents with differing structural characteristics. Data quality metrics are published in a variety of formats, including standards-style documents, table-based survey papers, and semi-structured research articles. A robust extraction system must therefore perform consistently across these heterogeneous inputs rather than being tailored to a single document type. The main objective was to determine how accurately and completely the system could extract data quality metrics and represent them in a structured form suitable for inclusion in a knowledge graph, assessed by comparing extracted outputs against a manually constructed gold-standard dataset of 123 annotated metrics.

The evaluation tests the following hypotheses, which correspond directly to the research questions defined in Section 1.3:

- **H1 (Extraction Accuracy — RQ1):** The proposed pipeline can correctly identify and extract the majority of data quality metrics from diverse document structures.
- **H2 (Structured Representation — RQ2):** Extracted metrics can be consistently represented as structured entities (`dqv:Metric`) with meaningful properties (label, definition, formula, dimension).
- **H3 (Practical Usefulness — RQ1):** Extracted outputs remain semantically useful even when not strictly exact under evaluation.
- **H4 (Knowledge Graph Queryability — RQ2):** The resulting RDF knowledge graph supports meaningful structured retrieval of metric knowledge by source, dimension, and metric detail.

To test these hypotheses, evaluation focused on four aspects:

- Extraction performance using precision, recall, and F1-score (H1).
- Property-level accuracy for key metric attributes (H1, H2).

- Partial match analysis to assess practical usefulness (H3).
- A separate SPARQL queryability experiment to demonstrate structured retrieval from the knowledge graph (H4).

6.2 Gold Standard and Evaluation Data

The system was evaluated against a manually constructed gold-standard dataset containing 123 annotated data quality metrics. The dataset was created by the author through manual extraction and annotation of metrics from three source documents with differing structural characteristics: a standards-style document (ETSI), a survey-based table source (Zaveri et al., 2015), and a semi-structured research paper (IEEE).

Annotation involved identifying individual metric definitions and extracting structured fields including metric label, definition, formula, and dimension. To ensure consistency, minor formatting artefacts introduced by PDF extraction (e.g., hyphenation across line breaks) were normalised during annotation.

The gold-standard dataset is available as part of the project repository: [<willinsertmylinkhere>](#).

- **Development and identification of Data Quality Metrics [4]:** a structured standards-style document,
- **Quality assessment for Linked Data: A Survey [3]:** a survey-style source with table-based metric presentation,
- **A Systematic Survey of Data Value: Models, Metrics, Applications and Research Challenges [5]:** a semi-structured research paper.

The extraction pipeline produced 119 records across the three evaluated sources. These outputs were generated automatically by the system and compared against the manually constructed gold-standard dataset containing 123 annotated metrics.

This near one-to-one correspondence between extracted records and gold-standard entries enables direct comparison, while still allowing analysis of false negatives (missed metrics) and partially correct extractions.

During gold-standard construction, a consistent set of annotation rules was applied to ensure that evaluation focused on semantic correctness rather than surface-level lexical or formatting variation. The gold standard was created manually by identifying individual metric definitions and extracting structured fields (MetricLabel, Definition, Formula, Dimension) according to a fixed schema aligned with the system output.

The following normalisation rules were applied consistently across all sources:

- **PDF artefacts:** Line-break hyphenation (e.g., “con-sidered” → “considered”), inconsistent whitespace, and merged words introduced by PDF extraction were corrected.
- **Label normalisation:** Parenthetical abbreviations embedded in labels (e.g., “Attribute Completeness (C_{Aj})”) were removed from the label field; the formula component was captured separately.

-
- **Definition boundaries:** Definitions were recorded to include only the prose description of the metric, excluding footnotes, continuation text, or content from adjacent sections.
 - **Formula capture:** Formulas were recorded where explicitly stated. Where no formal expression was present, the field was recorded as null rather than inferred.
 - **Dimension assignment:** Dimensions were recorded as stated in the source where available. For sources using non-DQV taxonomies (e.g., the IEEE source, which uses economic value categories), dimensions were mapped to the nearest DQV concept and flagged as inferred.
 - **Duplicate entries:** Where the same metric appeared in multiple tables within a source, each occurrence was preserved as a separate gold standard record, reflecting the document's organisation.

These rules were applied consistently across all three sources. Transformations were only permitted where they did not alter the semantic meaning of the text, ensuring that evaluation reflects the correctness of extracted knowledge rather than sensitivity to superficial formatting differences.

The gold standard was refined iteratively during evaluation. Early evaluation against a partial gold standard underestimated system precision because valid extracted metrics were incorrectly counted as false positives when they were absent from the incomplete gold standard. The final reported results use the completed and normalised 123-record gold standard. A summary of the score progression during this refinement process is provided in [Appendix A](#).

6.3 Experimental Setup

The evaluation compares two datasets derived from the same input documents: (i) a manually constructed gold-standard dataset and (ii) the output of the automated extraction pipeline.

The input consists of three documents with varying structural characteristics. The output of the pipeline is a structured representation of extracted metrics, which is then compared against the gold standard at the level of individual metric instances.

The pipeline took raw document text as input and produced structured CSV records containing four properties per metric instance (MetricLabel, Definition, Formula, Dimension), which were subsequently mapped to RDF and loaded into GraphDB.

All evaluation steps, including gold-standard construction and partial match classification, were performed by the author using predefined annotation and evaluation criteria.

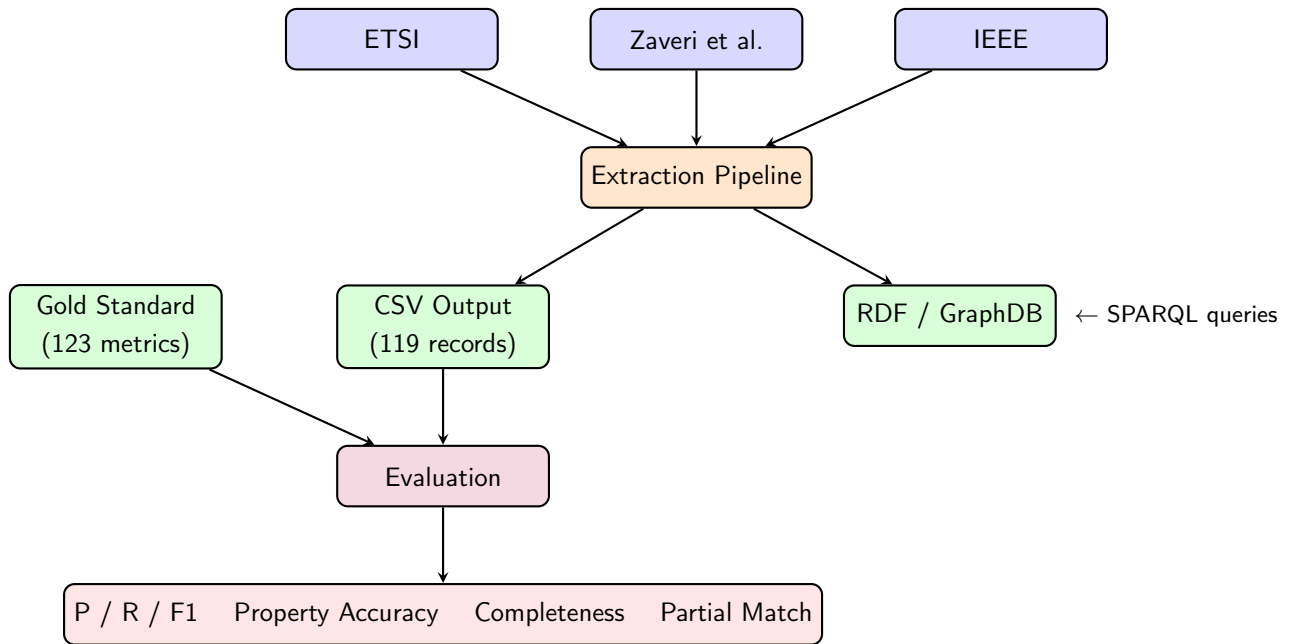


Figure 6.1: Experimental setup: data flow from source documents through the extraction pipeline to evaluation against the gold standard.

6.4 Experimental Procedure

The evaluation was conducted as a structured process consisting of the following steps:

1. **Data selection:** Three source documents with differing structures (ETSI, Zaveri, IEEE) were selected.
2. **Gold standard construction:** A manually annotated dataset of 123 metrics was created following the rules defined in Section 6.2.
3. **Automated extraction:** The extraction pipeline was applied to the same source documents, producing 119 extracted metric instances.
4. **Record alignment:** Extracted outputs were matched to corresponding gold-standard entries.
5. **Evaluation:** Performance was assessed using:
 - Precision, recall, and F1-score (exact and lenient),
 - Property-level accuracy,
 - Record completeness,
 - Partial match classification.
6. **Analysis:** Errors and mismatches were analysed to identify recurring failure modes and system limitations.

6.5 Evaluation Metrics Definition

The evaluation metrics used in this study are defined as follows:

Precision measures the proportion of extracted metrics that are correct with respect to the gold standard:

$$\text{Precision} = \frac{TP}{TP + FP}$$

where TP (true positives) are correctly extracted metrics and FP (false positives) are incorrectly extracted metrics.

Recall measures the proportion of gold-standard metrics that were successfully extracted:

$$\text{Recall} = \frac{TP}{TP + FN}$$

where FN (false negatives) are metrics present in the gold standard but not extracted by the system.

F1-score is the harmonic mean of precision and recall:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Property-Level accuracy measures correctness at the level of individual extracted attributes (MetricLabel, Definition, Formula, Dimension), computed separately for each field as the proportion of extracted records where that field matches the gold standard.

Record completeness measures the proportion of expected fields successfully extracted for each metric, averaged across all records.

Partial match analysis categorises extracted records into three classes:

- Exact: all fields match the gold standard,
- Partial: the correct metric is identified but one or more fields are incomplete or slightly incorrect,
- Incorrect: the metric is missing or incorrectly extracted.

SPARQL query demonstration assesses whether the populated knowledge graph supports meaningful structured retrieval through representative queries over source, dimension, and metric detail.

This multi-level evaluation provides a more informative assessment than a single binary correctness measure, particularly for an information extraction task where outputs may still be useful despite small errors, an approach consistent with evaluation frameworks used in recent LLM-based knowledge graph construction work [14].

6.6 Evaluation Methodology

Record-level performance was assessed at the level of individual metric instances (i.e., dqv:Metric entities in the knowledge graph) using precision, recall, and F1-score under both exact and lenient

matching conditions. These metrics were computed by comparing extracted outputs against the manually constructed gold-standard dataset (Section 6.2).

Precision, recall, and F1-score provide a summary view of extraction performance in terms of correctness and completeness. To capture more fine-grained aspects of system behaviour, additional analyses were conducted, including Property-Level accuracy (Section 6.6.2), record completeness (Section 6.6.3), and partial match analysis (Section 6.6.4).

6.6.1 Exact and Lenient Matching

Evaluation was performed under two matching conditions.

Exact matching required strict agreement between the extracted metric instance and the corresponding gold-standard entry. In this context, a “record” refers to a single data quality metric represented as a structured set of fields (MetricLabel, Definition, Formula, Dimension), corresponding to a dqv:Metric entity in the knowledge graph.

For example, the metric *Attribute Completeness* would be represented as a record containing its label, a textual definition, an associated formula (if present), and a dimension assignment (e.g., Completeness). Exact matching therefore requires that all fields align with the gold-standard entry with no significant differences.

Lenient matching allowed semantically correct outputs with controlled deviations from the gold standard to be counted as correct. In this context, leniency was defined using a fixed set of rules applied consistently across all evaluations.

A match was considered leniently correct if:

- **Formatting differences:** Differences in casing, whitespace, or punctuation did not affect the underlying content.
- **PDF artefacts:** Variations introduced by document formatting (e.g., hyphenation across line breaks such as “con-/nsidered”) were ignored following the normalisation rules defined in Section 6.2.
- **Definition variation:** Minor truncation or additional trailing text was permitted, provided the core definition of the metric was correctly captured.
- **Dimension equivalence:** Dimension labels were accepted if they were semantically equivalent or mapped to the same DQV-aligned concept (e.g., *reliability* vs. *consistency*).

All lenient matching decisions were determined by applying these predefined rules, rather than ad hoc judgement, ensuring consistency across the evaluation.

Outputs that introduced incorrect information, omitted core meaning, or referred to a different metric entirely were not accepted under lenient matching.

This distinction was particularly important because PDF-based inputs introduce formatting noise, and source documents do not always use a common dimension taxonomy. Lenient matching therefore provides a more realistic measure of practical usefulness while maintaining a consistent evaluation framework.

6.6.2 Property-Level accuracy

In addition to record-level matching, the system was evaluated at the level of individual properties of each metric instance. The following properties were assessed:

- **Metric label**
- **Definition**
- **Formula**
- **Dimension**

This made it possible to identify which parts of the extraction pipeline were strongest and which remained the main source of error.

6.6.3 Record Completeness

Record completeness was measured to quantify how fully each correctly matched metric instance was captured. In this context, a record corresponds to a single `dqv:Metric` entity represented by a set of properties.

Completeness was computed as the proportion of key properties correctly extracted for each matched metric. The following properties were considered:

- **Metric label** (mandatory)
- **Definition** (mandatory)
- **Formula** (optional, where present in the source)
- **Dimension** (optional, where explicitly stated or inferred)

Mandatory properties (label and definition) were required for a metric to be considered valid, as they capture the core identity and meaning of the metric. Optional properties were included in completeness calculations only when present in the gold standard.

Record completeness was averaged across all matched metric instances.

6.6.4 Partial Match Analysis

An additional match quality analysis was performed to distinguish between fully correct, partially correct, and incorrect extractions.

Each gold-standard metric instance was classified manually by the author into one of three categories using a fixed set of criteria:

- **Exact:** All properties (MetricLabel, Definition, Formula, Dimension) match the gold standard, allowing only trivial formatting differences (e.g., casing, whitespace, or punctuation).

- **Partial:** The extracted metric corresponds to the correct metric instance (i.e., the MetricLabel identifies the same underlying concept), but one or more properties are incomplete or deviate from the gold standard. This includes:
 - Definition boundary issues (e.g., truncation or additional trailing text),
 - Minor formatting artefacts introduced by PDF extraction,
 - Dimension mismatches where the extracted value is semantically equivalent or mapped to the same DQV-aligned concept.
- **Incorrect:** The extracted output does not correspond to the gold-standard metric, introduces incorrect information, or the metric is entirely missing.

Classification was performed deterministically by the author based on these criteria rather than ad hoc judgement, ensuring consistency across all evaluated records.

6.7 Results

6.7.1 Precision, Recall and F-Measure

Table 6.1 summarises the overall extraction performance under exact and lenient matching conditions.

Match Type	Precision	Recall	F1-score
Exact	0.908	0.878	0.893
Lenient	1.000	0.967	0.983

Table 6.1: Overall extraction performance under exact and lenient matching.

The exact results indicate strong overall extraction performance, with an F1-score of 0.893. Under lenient matching, performance rises further to an F1-score of 0.983, showing that many outputs classified as imperfect under strict evaluation were still practically useful.

The final scores reflect an iterative gold standard refinement process; see Appendix A for the full progression.

6.7.2 Property-Level Accuracy

Table 6.2 shows performance for individual extracted fields.

The strongest components of the system were metric label extraction, definition extraction, and formula extraction. Metric labels were nearly always identified correctly, while formula extraction was also highly reliable. This indicates that the extraction pipeline was effective at capturing the core content of each metric.

Dimension classification was the weakest property, achieving 52.1% exact accuracy and 60.5% lenient accuracy, compared with 98.3% for metric labels, 92.4% for definitions, and 99.2% for formulas. This lower performance reflects the fact that dimensions were often inferred heuristically rather than explicitly stated in the source text. In addition, some sources did not use DQV-style dimensions, making exact agreement difficult even when the extracted metric itself was correct.

Field	Accuracy
Metric label (exact)	0.983
Metric label (lenient)	1.000
Definition (exact)	0.924
Definition (lenient)	0.992
Formula	0.992
Dimension (exact)	0.521
Dimension (lenient)	0.605

Table 6.2: Property-level extraction accuracy.

6.7.3 Record Completeness

Average record completeness was 0.935 under exact scoring and 0.977 under lenient scoring. This indicates that, on average, most matched records contained nearly all of the expected fields, even when minor imperfections were present.

This result is important because it shows that the system does not merely identify metric labels in isolation. In most cases, it also captures the surrounding descriptive information needed for downstream use in a knowledge graph.

6.7.4 Per-Source Results

Performance varied slightly across the three evaluated sources, as shown in Table 6.3.

Source	Precision	Recall	F1	Exact	Partial	Unmatched	GoldStandard
ETSI	1.000	1.000	1.000	12	0		0
Zaveri	0.855	0.855	0.855	59	10		0
IEEE	0.974	0.881	0.925	37	1		4

Table 6.3: Per-source extraction results under exact scoring. Partial matches are shown separately and are not counted as true positives in the precision, recall, and F1-score values.

Under exact scoring, ETSI achieved perfect performance with precision, recall, and F1-score all equal to 1.000. Zaveri achieved precision, recall, and F1-score of 0.855, with the gap from perfect performance attributable to 10 partial matches caused by definition boundary issues. IEEE achieved precision of 0.974, recall of 0.881, and F1-score of 0.925, with lower recall reflecting four unmatched gold-standard entries. Two of these were caused by duplicate metric occurrences in the source tables, while the remaining two correspond to metrics the pipeline did not extract, likely due to definition boundary or PDF parsing issues.

The high precision values across all sources reflect the conservative nature of the extraction strategies used. ETSI was processed using deterministic rule-based parsing over a structured standards-style document, while Zaveri used a table-specific extraction strategy with known metric-name anchors from the source table. These choices reduce the likelihood of spurious extracted candidates for these specific documents. The precision values should therefore be interpreted as evidence of correct and conservative extraction for these target documents, rather than as a claim of generalized out-of-domain precision. This limitation is discussed further in Section 6.12.

6.7.5 Partial Match Analysis

A more nuanced view of output quality is provided by the match quality analysis shown in Table 6.4.

Source	Exact	Partial	Missed	Total
ETSI	12	0	0	12
Zaveri	59	10	0	69
IEEE	37	1	4	42
Overall	108	11	4	123

Table 6.4: Match quality analysis across all gold-standard records.

Overall, 108 records (87.8%) were exact matches, 11 (8.9%) were partial matches, and 4 (3.3%) were classified as missed. This confirms that the vast majority of outputs were either fully correct or still useful with only minor issues.

This is especially important because strict precision and recall alone do not adequately capture the practical value of outputs that are semantically correct but imperfectly formatted. In the context of knowledge curation, a partially correct metric is often still far more useful than a completely missing one.

6.8 Error Analysis

Table 6.5 summarises the observed error types and their frequency across all evaluated records. Table 6.6 provides representative examples to clarify how these categories affected evaluation.

Error Type	Count	Source(s)
Definition boundary / bleed	11	Zaveri, IEEE
Duplicate / granularity mismatch	2	IEEE
Missing extraction	2	IEEE
Other (semantic / label / typo)	5	Zaveri, IEEE
Dimension mismatch (heuristic)	0	(handled / excluded)
Total affected records	20	

Table 6.5: Error type frequency across all evaluated records classified as partial or incorrect.

These findings indicate that most remaining errors arise from representation and boundary-related issues rather than complete failure to identify the underlying metric. This is consistent with the partial match analysis, which shows that many non-exact outputs remain semantically useful.

The main remaining errors fell into a small number of recurring categories.

6.8.1 PDF Formatting Artefacts

A significant source of noise came from PDF extraction artefacts such as line-break hyphenation, broken spans, and loss of table structure. These artefacts affected both extraction and evaluation. For example, a definition or label could be semantically correct but still differ textually from the gold standard because of formatting introduced by the original PDF.

Error Type	Example	Evaluation Impact
Definition boundary / bleed	The metric <i>Freshness of datasets based on their data source</i> was correctly identified, but the extracted definition included surrounding text from adjacent sections.	Classified as a partial match because the underlying metric was correct, but the definition property contained additional irrelevant text.
Duplicate / granularity mismatch	The IEEE metrics <i>Market price</i> and <i>Rival access loss</i> appeared in more than one source table.	Counted as unmatched gold-standard entries where the gold standard preserved both occurrences but the extractor deduplicated by metric label.
Missing extraction	Two IEEE metrics present in the gold standard were not returned by the pipeline, likely due to definition boundary detection failures or loss of table structure during PDF parsing.	Counted as false negatives because no corresponding extracted metric instance was available for alignment.

Table 6.6: Examples of observed error types and their evaluation impact.

6.8.2 Definition Boundary Detection

In some cases, the correct metric definition was extracted but was followed by irrelevant surrounding text from the source page. This *definition bleed* reduced exact-match performance even though the extracted output remained directionally correct. These cases were treated as partial matches rather than full errors. Figure 6.2 illustrates a typical case from the Zaveri source.

Definition Bleed Example (Zaveri)

Gold standard:
“Detecting freshness of datasets based on their data source by measuring the distance between last modified time of the data source and last modified time of the dataset.”

Extracted:
“Detecting freshness of datasets based on their data source by measuring the distance between last modified time of the data source and last modified time of the dataset QN — T3: using trust ontologies that assigns trust values that can be transferred from known to unknown data using: (i) content-based methods ... [continues for several sentences]”

The correct definition ends after “dataset”. All subsequent text belongs to an adjacent section and was incorrectly captured due to a missing boundary signal in the extracted PDF text.

Figure 6.2: Example of definition bleed where trailing text from an adjacent section was incorrectly included in the extracted definition.

6.8.3 Dimension Inference and Taxonomy Differences

The IEEE source does not natively organise metrics using DQV dimensions. Instead, it groups them under economic value categories: Financial, Usage, Utility, and Content-Uniqueness. These categories do not map cleanly onto the DQV-oriented taxonomy used in the knowledge graph. As a result, the extractor was unable to resolve IEEE dimension values and output `unknown`

for all 38 IEEE metric instances. Dimension fields for IEEE were therefore excluded from dimension accuracy scoring. Figure 6.3 illustrates this.

Taxonomy Mismatch Example (IEEE)

Source dimension (gold standard): Utility

Extracted dimension: unknown

The extractor could not resolve IEEE's economic value categories (Financial, Usage, Utility, Content-Uniqueness) to DQV-aligned terms. Dimension fields for all IEEE metrics are therefore output as unknown and excluded from dimension accuracy scoring.

Figure 6.3: Example of taxonomy mismatch where IEEE economic value categories cannot be resolved to DQV-style quality dimensions by the extractor.

6.8.4 Duplicate Metrics in Source Tables

A small number of false negatives arose from duplicate metric appearances in the IEEE source. In particular, *Market price* and *Rival access loss* appeared in more than one table, under different source contexts such as Utility and Financial. The gold standard preserved these as separate entries because each occurrence reflects a distinct position in the source document with a different categorical context.

The extractor, however, deduplicated records using only the normalised metric label and source, retaining only the first occurrence encountered. This means that the underlying metric label was identified, but the system failed to preserve all source-level occurrences. These cases are therefore best understood as an over-aggressive deduplication error rather than a complete failure to recognise the metric. A corrective fix would incorporate dimension into the deduplication key; however, since the IEEE extractor outputs *unknown* for all dimension values, this would require resolving the taxonomy mismatch described in Section 6.8.3 first. Both limitations are acknowledged as directions for future work.

Duplicate Granularity Example (IEEE)

Gold standard entry 1 (Table: Utility):
Label: Market price Dimension: Utility

Gold standard entry 2 (Table: Financial):
Label: Market price Dimension: Financial

Extracted (deduplicated):
Label: Market price Dimension: unknown

The extractor retained only one occurrence of Market price, as deduplication was performed by label alone. The second gold-standard entry was therefore unmatched, contributing to the IEEE false negative count.

Figure 6.4: Example of over-aggressive deduplication where the extractor produced one record for a metric appearing in two distinct source-table contexts.

6.9 Knowledge Graph Queryability Experiment

This section reports a separate queryability experiment designed to assess whether the extracted metrics, once represented as RDF and loaded into GraphDB, can support meaningful structured retrieval. Unlike the extraction evaluation, this experiment does not assess precision or recall against the gold standard. Instead, it tests whether the populated knowledge graph can answer representative information needs using SPARQL.

6.9.1 Hypothesis

- **H4 (Knowledge Graph Queryability — RQ2):** The resulting RDF knowledge graph supports meaningful structured retrieval of metric knowledge by source, dimension, and metric detail.

6.9.2 Experimental Setup

The experiment was conducted on the live GraphDB repository populated with RDF generated from the extraction pipeline. The input was the RDF/Turtle output produced from the 119 extracted metric records via R2RML mapping. The output consisted of SPARQL query results returned by GraphDB.

Three representative SPARQL queries were designed to reflect retrieval tasks a user might perform over a data quality metric knowledge base:

- **Q1: Metrics by source** — checks whether metrics can be retrieved and grouped by provenance.
- **Q2: Metrics by dimension** — checks whether the DQV-aligned dimension structure is queryable.
- **Q3: Metric detail retrieval** — checks whether a user can retrieve metric labels, definitions, and formulas for a named metric.

6.9.3 Procedure

1. Extracted metric records were converted to RDF/Turtle via R2RML mapping.
2. The RDF output was loaded into GraphDB.
3. Three representative SPARQL queries were executed against the repository.
4. Query results were inspected to confirm whether the expected source, dimension, and metric-detail information could be retrieved.

6.9.4 Results

Table 6.7 summarizes the three queries executed and their retrieval purpose. The corresponding SPARQL outputs are shown in Listings 6.1–6.2 and Table 6.8.

Query	Retrieval Task	Purpose
Q1	Metrics by source	Confirms provenance-based retrieval
Q2	Metrics by dimension	Confirms DQV-style dimension navigation
Q3	Metric detail by label	Confirms detailed metric lookup

Table 6.7: Representative SPARQL queries used to demonstrate knowledge graph queryability.

The first query retrieved metric counts grouped by source, confirming that extracted records are distinguishable by provenance after RDF transformation.

source	count
-----	-----
Zaveri	69
IEEE	38
ETSI	12

Listing 6.1: Q1 — Metric counts by source.

The second query retrieved populated dimensions and associated metric counts, demonstrating that the DQV-aligned structure supports dimension-oriented navigation.

dimLabel	count
-----	-----
Consistency	15
Semantic accuracy	12
Completeness	7
Trustworthiness	7
Understandability	6
Conciseness	5
Availability	5
Interpretability	4
Performance	4
Syntactic validity	3
Licensing	3
Relevancy	2
Interoperability	2
Versatility	2
Representational-conciseness	2
Timeliness	2
Security	2

Listing 6.2: Q2 — Metric counts by dimension.

The third query retrieved detailed metric records including labels, definitions, and formulas, confirming that structured details remain retrievable after transformation into RDF.

label	definition	formula
Attribute Completeness	The completeness of a specific attribute A_j is given by the ratio of present values for that attribute to the total number of records.	$C_{A_j} = \frac{\sum_{i=1}^N I_{ij}}{N}$
Dataset Completeness	The overall completeness of the dataset, considering all N records and M attributes...	$C_D = \frac{\sum_{j=1}^{M_{req}} \sum_{i=1}^N I_{ij}}{N \times M_{req}}$
Record Completeness	The completeness of a specific record R_i ...	$C_{R_i} = \frac{\sum_{j \in Req} I_{ij}}{K}$
Interlinking completeness	No. of instances in the dataset that are interlinked...	NULL
Population completeness	No. of real-world objects are represented...	NULL

Table 6.8: Q3 — Metric detail retrieval (excerpt).

6.9.5 Discussion

The results provide evidence for H4 by demonstrating that the RDF knowledge graph enables structured retrieval across all three tested query types. Q1 confirms that extracted metrics remain distinguishable by source provenance after RDF transformation. Q2 demonstrates that the DQV-aligned dimension structure supports navigation, although the usefulness of this query depends on the quality of dimension inference — dimension-oriented retrieval is less complete where source documents do not use DQV-style dimensions, as is the case for the IEEE source. Q3 demonstrates detailed metric lookup, confirming that labels, definitions, and formulas remain retrievable after transformation into RDF.

This experiment should be interpreted as a queryability demonstration rather than a full competency-question evaluation. A more extensive future evaluation could define a formal set of competency questions and measure query success against those requirements.

6.10 Discussion of Evaluation Results

The extraction evaluation provides support for H1–H3. H1 is supported by the strong overall extraction performance, with an exact F1-score of 0.893 and a lenient F1-score of 0.983, demonstrating that the pipeline correctly identifies the majority of metrics across diverse document structures.

H2 is supported by high property-level accuracy for metric labels (98.3%), definitions (92.4%), and formulas (99.2%), confirming that extracted metrics contain sufficient structured information for knowledge graph representation. The main exception is dimension classification, which achieved lower accuracy due to heuristic inference and source-specific taxonomy differences.

H3 is supported by the partial match analysis, which shows that 87.8% of gold-standard metric instances were exact matches and a further 8.9% remained semantically useful as partial matches. This indicates that many non-exact outputs were still useful despite minor issues such as definition boundary errors or formatting artefacts.

H4 is addressed separately in the knowledge graph queryability experiment (Section 6.9), which demonstrates that the RDF representation supports structured retrieval by source, dimension, and

metric detail.

The results also support the design choice of using a hybrid extraction pipeline. Structured rule-based extraction worked well for standards-style inputs, while source-specific table handling improved robustness on survey-style documents. More general NLP-based extraction provided flexibility for less regular text. The variation in source structure justifies the use of different extraction strategies rather than a single uniform method.

As discussed in Section 6.11, comparison with large language models highlights a key trade-off: while LLMs can generalise well to extraction tasks, they exhibit sensitivity to prompt formulation, over-generation from incomplete source text, and occasional context leakage, whereas the proposed pipeline provides more controlled and reproducible outputs.

6.11 LLM-Based Extraction Baseline

To contextualise the pipeline's outputs, a lightweight baseline evaluation was conducted using two general-purpose large language models: Claude Opus and ChatGPT. Each model was given the same raw document text and prompted to extract metrics with the fields `MetricLabel`, `Definition`, `Formula`, and `Dimension`. The goal was not to produce a formal benchmark, but to observe how LLMs handle the same extraction task across structurally different inputs.

This experiment was conducted using extracted text rather than the original PDF files, to ensure consistency of input and isolate extraction performance from document parsing effects.

Prompt Compliance and Granularity

Claude produced correctly structured output across all three sources without requiring prompt refinement. ChatGPT initially misinterpreted the ETSI document, returning four section-level entries — one per quality dimension — rather than the individual named sub-metrics within each section. After re-prompting with an explicit instruction to extract each individually named metric separately, ChatGPT produced outputs consistent with the expected granularity and structure defined by the gold standard.

Over-Generation

Both models extracted metrics from sections of the ETSI document that are incomplete in the source draft, specifically sections covering Timeliness, Consistency, and Integrity, which contain only placeholder formulae and no authored definition text. Claude returned 31 metrics and ChatGPT returned 30 after re-prompting, compared to the 12 gold-standard entries. The rule-based pipeline produced no output for these sections, as the quality gate enforces a minimum definition length and requires prose context. This illustrates a key behavioural difference: LLMs infer plausible content from partial input, whereas the deterministic pipeline only extracts what is explicitly present in the source. This behaviour reflects the generative nature of LLMs, which prioritise plausibility over strict adherence to source completeness.

Cross-Source Consistency

On the Zaveri source, both models performed well. A minor label formatting difference was observed: Claude preserved the document's short-code labels (e.g. A1, A2), while ChatGPT nor-

malised these to full descriptive names. This is a presentational difference rather than a content error. For the IEEE source, ChatGPT's output included metrics that do not appear in the source document, suggesting context leakage or confusion when processing longer or more complex inputs.

It should be noted that LLM outputs were not evaluated against the gold standard using the precision, recall, and property-level methodology applied to the pipeline. A direct comparison would require controlled prompt design, multiple runs, and output normalisation, which were outside the scope of this project.

Summary

The LLM baseline demonstrates that general-purpose models can extract data quality metrics with reasonable accuracy given a well-formed prompt. However, several failure modes were observed: sensitivity to prompt wording (ChatGPT, ETSI attempt 1), over-generation from incomplete source sections (both models, ETSI), and occasional context leakage on longer inputs (ChatGPT, IEEE). The proposed pipeline avoids these failure modes through explicit structural parsing and quality constraints, at the cost of reduced flexibility when handling incomplete or ambiguous source content.

6.12 Evaluation Limitations

The evaluation has several limitations. First, it depends on the quality and completeness of the manually constructed gold standard. Although care was taken to normalise formatting artefacts and ensure broad coverage, the gold standard still reflects modelling decisions about what counts as a distinct metric record — for example, whether duplicate metric appearances in source tables should be preserved as separate entries.

Second, exact matching may understate practical performance by penalising outputs that are semantically correct but textually imperfect. This was partially addressed through lenient scoring and partial match analysis, but some judgement calls in classification remain unavoidable, particularly for borderline partial matches. Third, the evaluation was conducted on a fixed set of three source documents selected to represent different structural types. While this provides diversity, it limits the generalisability of the results to unseen document types or domains beyond data quality literature.

Despite these limitations, the evaluation provides sufficient evidence to assess the system's effectiveness across structurally diverse sources and to identify its principal failure modes, supporting the conclusions drawn in this chapter.

Chapter 7: Project Workplace

The system was developed using a combination of Python-based processing and semantic web technologies. The project was implemented and tested on a macOS local development environment.

The extraction pipeline was implemented in Python, using spaCy [15] for sentence segmentation and linguistic processing, along with regular expressions for rule-based extraction and fallback handling. Document parsing was performed using PyMuPDF [16] for PDF text extraction and `python-docx` for DOCX handling.

The structured output was stored in CSV format and transformed into RDF using the R2RML reference implementation [10]. The resulting RDF data was represented in Turtle format and loaded into GraphDB [17], which was used as the primary triple store for querying and validation.

All development and testing were carried out locally, with GraphDB used to execute SPARQL queries and verify the structure and completeness of the generated knowledge graph. Reproducibility instructions, including exact CLI commands for running the pipeline and generating RDF, are provided in Appendix B.

This environment enabled rapid iteration across the extraction and transformation pipeline while supporting inspection and validation of intermediate outputs.

Chapter 8: Summary and Conclusions

This project investigated the problem of extracting and structuring data quality metric definitions from academic papers, standards documents, and table-heavy sources. Existing metric knowledge is widely dispersed across academic literature and standards documents, making it difficult to discover, compare, and reuse in practice. The primary objective of this work was to design and evaluate a system capable of automatically extracting this information and representing it in a structured, machine-readable form.

To address this challenge, a hybrid extraction pipeline was developed, combining rule-based methods, table reconstruction techniques, and NLP-based processing. This approach was motivated by the structural diversity of the input sources, where no single extraction strategy was sufficient. Extracted data was transformed into a structured representation and mapped to a knowledge graph aligned with W3C vocabularies, including the Data Quality Vocabulary (DQV), SKOS, and Dublin Core. The resulting graph was stored in GraphDB and queried using SPARQL.

The system was evaluated using a manually constructed gold-standard dataset consisting of 123 annotated metrics drawn from three sources with varying document structures. Evaluation results demonstrated strong overall performance, with an exact F1-score of 0.893 and a lenient F1-score of 0.983. Property-level analysis showed that metric labels (98.3%), definitions (92.4%), and formulas (99.2%) were extracted with high accuracy, while dimension classification remained the most challenging component due to reliance on heuristic inference and differences in source taxonomies.

A key finding of the evaluation is that strict binary metrics such as precision and recall do not fully capture the practical usefulness of extracted outputs. Partial match analysis showed that many non-exact outputs were still semantically useful, highlighting the importance of multi-level evaluation in information extraction tasks.

In addition to extraction performance, the project demonstrated that the resulting knowledge graph supports meaningful structured retrieval. SPARQL queries confirmed that metrics can be retrieved by source, dimension, and detailed content, showing that the system transforms extracted metric information into a queryable knowledge resource.

The LLM baseline comparison further highlighted a trade-off between flexibility and control. While large language models were capable of extracting data quality metrics with appropriate prompting, they showed sensitivity to prompt wording, over-generation from incomplete sections, and occasional context confusion. In contrast, the proposed pipeline provided more controlled and reproducible outputs, although with less flexibility on incomplete or ambiguous input.

Despite these strengths, several limitations were identified. The system is dependent on the quality of PDF text extraction, particularly for table-heavy documents where structural information is often lost. Dimension classification relies on heuristic mapping rather than authoritative labels, leading to systematic mismatches in some cases. Furthermore, differences in how source documents conceptualise metrics, such as economic value categories versus quality dimensions, introduce inconsistencies that cannot be fully resolved automatically.

Future work could address these limitations by incorporating more robust document parsing techniques, including layout-aware PDF processing or table extraction models. Dimension classification could be improved through supervised or semi-supervised approaches to better align source terminology with standard taxonomies. In addition, expanding the dataset to include a broader range

of documents would improve generalisability and enable more comprehensive evaluation.

Overall, the project provides affirmative answers to both research questions. RQ1 is supported by strong extraction performance across structurally diverse sources, and RQ2 is supported by the demonstration that a DQV-aligned knowledge graph enables meaningful structured retrieval through SPARQL. By combining a hybrid extraction strategy with standards-aligned knowledge graph representation, the system provides a practical approach to transforming fragmented metric definitions into a structured, queryable resource.

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Appendix A: Evaluation Score Progression

The final evaluation scores were reached through an iterative process of gold standard refinement. Table [A.1](#) shows how performance metrics evolved across key development stages.

Stage	P	R	F1	FP	FN	Change
Initial (60 gold)	0.395	0.783	0.525	61	2	Partial gold standard
Full gold (123)	0.790	0.764	0.777	14	6	Added all metrics
+ Hyphen fix	0.908	0.878	0.893	1	5	Cleaned GS artefacts
Final (stable)	0.908	0.878	0.893	0	4	Stable final state

Table A.1: Evaluation score progression across gold standard refinement stages.

The initial low precision of 0.395 reflected incomplete gold standard coverage rather than extraction failure. Valid extracted metrics were being penalised as false positives because the partial gold standard did not yet include all extractable metrics. After expanding to 123 metrics and correcting PDF hyphenation artefacts in the gold standard, scores stabilised at P=0.908, R=0.878, and F1=0.893.

Appendix B: Reproducibility and Execution Commands

B.1 Running the Extraction Pipeline

For readability, shortened input filenames are shown below; the full original filenames are preserved in the project repository.

```
python3 src/extractor.py \  
  --input "data/Inputs/DATA-00104180v004.docx" \  
  --source "ETSI"  
  
python3 src/extractor.py \  
  --input "data/Inputs/zaveri-survey.pdf" \  
  --source "Zaveri"  
  
python3 src/extractor.py \  
  --input "data/Inputs/ieee-data-value-survey.pdf" \  
  --source "IEEE"
```

B.2 Generating RDF

```
java -jar "../Tools/r2rml-master/target/r2rml-fat.jar" config.properties
```

Appendix C: SPARQL Queries

C.1 Q1: Metrics by Source

```
PREFIX dqv:      <http://www.w3.org/ns/dqv#>
PREFIX dcterms:  <http://purl.org/dc/terms/>

SELECT ?source (COUNT(?metric) AS ?count) WHERE {
  GRAPH <https://emmanuelawe51.github.io/dqkg/data/metrics> {
    ?metric a dqv:Metric ;
            dcterms:bibliographicCitation ?source .
  }
}
GROUP BY ?source
ORDER BY DESC(?count)
```

C.2 Q2: Metrics by Dimension

```
PREFIX dqv:      <http://www.w3.org/ns/dqv#>
PREFIX skos:      <http://www.w3.org/2004/02/skos/core#>

SELECT ?dimLabel (COUNT(?metric) AS ?count) WHERE {
  GRAPH <https://emmanuelawe51.github.io/dqkg/data/metrics> {
    ?metric a dqv:Metric ;
            dqv:inDimension ?dim .
  }
  GRAPH <https://emmanuelawe51.github.io/dqkg/vocab/ldqd> {
    ?dim skos:prefLabel ?dimLabel .
  }
}
GROUP BY ?dimLabel
ORDER BY DESC(?count)
```

C.3 Q3: Metric Detail Retrieval

```
PREFIX dqv:      <http://www.w3.org/ns/dqv#>
PREFIX skos:      <http://www.w3.org/2004/02/skos/core#>

SELECT ?label ?definition ?formula WHERE {
  GRAPH <https://emmanuelawe51.github.io/dqkg/data/metrics> {
    ?metric a dqv:Metric ;

```

```
        skos:prefLabel ?label ;
        dqv:inDimension <http://www.w3.org/2016/05/ldqd#completeness> .
    OPTIONAL { ?metric skos:definition ?definition }
    OPTIONAL { ?metric <https://emmanuelawe51.github.io/dqkg/formula> ?formu
}
}
ORDER BY ?label
```

Appendix D: LLM Baseline Prompt

Extract all data quality metrics from the following text.

For each metric, return:

- MetricLabel
- Definition
- Formula, if present
- Dimension, if mentioned

Please place the results within a csv file

...

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